

NEET-UG 2026 — Chemistry Categorization

Each of the 45 Chemistry questions (Q46–Q90) in the NEET-UG 2026 paper, mapped to its corresponding Unit / Subtopic in the official NEET-UG 2026 syllabus.

Sources: NEET-UG 2026 syllabus (NMC notification, Notice_20260108180635.pdf, Chemistry — 20 units) · Question paper + answer key + line-by-line explanations (questions on pp. 2–86, explanations on pp. 87–277).

Distribution across the syllabus units

Unit	Syllabus Unit	Q count	Question Numbers
1	Some Basic Concepts in Chemistry	1	Q63
2	Atomic Structure	2	Q47, Q82
3	Chemical Bonding and Molecular Structure	3	Q51, Q77, Q86
4	Chemical Thermodynamics	2	Q78, Q79
5	Solutions	2	Q74, Q75
6	Equilibrium	3	Q71, Q85, Q87
7	Redox Reactions and Electrochemistry	2	Q53, Q70
8	Chemical Kinetics	3	Q57, Q65, Q67
9	Classification of Elements and Periodicity in Properties	2	Q56, Q84
10	P-Block Elements	3	Q76, Q83, Q90
11	d- and f-Block Elements	3	Q54, Q64, Q68
12	Co-ordination Compounds	3	Q55, Q62, Q81
13	Purification and Characterisation of Organic Compounds	1	Q50
14	Some Basic Principles of Organic Chemistry	2	Q61, Q72
15	Hydrocarbons	3	Q49, Q59, Q89
16	Organic Compounds Containing Halogens	1	Q73
17	Organic Compounds Containing Oxygen	3	Q60, Q66, Q88
18	Organic Compounds Containing Nitrogen	1	Q80
19	Biomolecules	1	Q46
20	Principles Related to Practical Chemistry	4	Q48, Q52, Q58, Q69
Total		45	

Class-level split (NCERT)

- **Class XI** topics (Units 1–10 — Physical & general): ~23 questions
- **Class XII** topics (Units 11–20 — Inorganic, Organic, Practical): ~22 questions

Per-question categorization

Q#	Ans	Syllabus Unit	Subtopic	What the question tests (from the PDF explanation)
46	B	19 — Biomolecules	Nucleic acids — DNA / RNA secondary structure	DNA is double-stranded with bases A,G,C,T; RNA is single-stranded with A,G,C,U.
47	B	2 — Atomic Structure	Quantum numbers; orbital notation (n, l)	Match (n, l) pairs to s/p/d/f orbital notation using $l = 0-3 \Rightarrow s/p/d/f$.
48	A	20 — Principles Related to Practical Chemistry	Detection of extra elements — Lassaigne's test	Fusion with metallic Na converts C, N, S, X (halogens) in organics to their Na salts (NaCN, Na ₂ S, NaX).
49	D	15 — Hydrocarbons	Aromatic compounds — electrophilic substitution / chlorination	C ₆ H ₆ + 6Cl ₂ under anhyd. AlCl ₃ and UV gives different chlorinated products X and Y; count Cl atoms.
50	A	13 — Purification and Characterisation of Organic Compounds	Separation techniques — fractional distillation	Two miscible liquids X and Y with different boiling points — separated by fractional distillation.
51	C	3 — Chemical Bonding and Molecular Structure	VSEPR theory — shape and lone pairs	ClF ₃ with 5 electron pairs on Cl (3 BP + 2 LP) → T-shaped molecular geometry.
52	C	20 — Principles Related to Practical Chemistry	Detection of functional groups — phthalein dye test	Phthalic anhydride + active-H compound + conc. H ₂ SO ₄ → leuco base → coloured phthalein — confirms phenolic group.
53	A	7 — Redox Reactions and Electrochemistry	Faraday's laws of electrolysis — mass deposited	Mass of Cu deposited at cathode from given current and time using $m = (I \cdot t \cdot M)/(n \cdot F)$.
54	A	11 — d- and f-Block Elements	Transition metal / oxide catalysis in industrial processes	Match: V ₂ O ₅ → Contact process (H ₂ SO ₄ from SO ₂); Fe → Haber process (NH ₃ from N ₂ /H ₂).
55	B	12 — Co-ordination Compounds	Geometry of complexes (square planar, octahedral, tetrahedral)	[Pt(NH ₃) ₂ Cl ₂] square planar; [Co(NH ₃) ₆]Cl ₃ octahedral; identify geometry of each complex.
56	A	9 — Classification of Elements and Periodicity in Properties	Atomic and ionic radii — size comparison	Cations smaller than parents; for isoelectronic Mg ²⁺ vs Al ³⁺ greater Z gives smaller size.
57	C	8 — Chemical Kinetics	Arrhenius equation — activation energy	Given $\ln k = 14.34 - 1.25 \times 10^4/T$, compare to $\ln A - E_a/RT \rightarrow E_a = 1.25 \times 10^4 \times R$.
58	C	20 — Principles Related to Practical Chemistry	Titrimetric exercises — acid-base indicators	NaOH (alkaline, phenolphthalein pink) titrated with oxalic acid → colour change pink → colourless.
59	A	15 — Hydrocarbons	Reactions of alkanes — steam reforming of methane	CH ₄ + H ₂ O —(Ni, 1273 K)→ CO + 3H ₂ (syngas).
60	A	17 — Organic Compounds Containing Oxygen	Functional-group conversions — benzene to phenolsulfonic acid; carboxylic acid to alcohol	Reagent matching: oleum / NaOH Δ / H ⁺ for benzene → p-hydroxybenzenesulfonic acid; reductions to convert CH ₃ COOH → CH ₃ CH ₂ OH.
61	A	14 — Some Basic Principles of Organic Chemistry	Isomerism — metamerism	Metamers: same molecular formula + same functional group, different alkyl groups across the functional group (e.g. ethers like CH ₃ OC ₃ H ₇ vs C ₂ H ₅ OC ₂ H ₅).
62	D	12 — Co-ordination Compounds	Isomerism in coordination compounds (geometrical, optical, ionisation, solvate)	[Pt(NH ₃) ₂ Cl ₂] cis/trans (geometrical); [Co(en) ₃] ³⁺ optical; match each complex to its isomerism type.
63	B	1 — Some Basic Concepts in Chemistry	Mole concept — Avogadro's number, atom counting	Number of H atoms in given mass of urea via moles × N _A × (H atoms per formula unit).
64	D	11 — d- and f-Block Elements	Magnetic moment of transition metal ions (spin-only)	Ti ²⁺ is 3d ² → n = 2 unpaired → $\mu_s = \sqrt{n(n+2)} = \sqrt{8}$ BM.
65	D	8 — Chemical Kinetics	Order of reaction from concentration–time plot	Linear decrease in [R] vs t at constant slope → zero-order; rate = k, independent of [R].

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66	A	17 — Organic Compounds Containing Oxygen	Aldehydes and ketones — chemical tests (2,4-DNP, Fehling, Tollens)	C ₈ H ₈ O compound: 2,4-DNP +ve and Fehling –ve → ketone or aromatic aldehyde — identify P and Q from given tests.
67	A	8 — Chemical Kinetics	Order and units of the rate constant	Match: zero-order → mol L ⁻¹ s ⁻¹ ; 1st → s ⁻¹ ; 2nd → mol ⁻¹ L s ⁻¹ ; 3rd → mol ⁻² L ² s ⁻¹ .
68	C	11 — d- and f-Block Elements	Lanthanoid oxidation states — stability of +4 in Ce	Ce ([Xe] 4f ¹ 5d ¹ 6s ²) → Ce ^{IV} has empty 4f configuration which is energetically favourable.
69	C	20 — Principles Related to Practical Chemistry	Qualitative anion analysis — acetate (CH ₃ COO ⁻)	Colourless vinegar-smelling acidic vapours on adding dil. H ₂ SO ₄ → acetate ion (acetic acid liberated).
70	D	7 — Redox Reactions and Electrochemistry	Nernst equation — EMF of half-cell	H ₂ / H ⁺ electrode: $E = E^\circ - (2.303RT/nF) \log([H^+]^2 / p_{H_2})$.
71	A	6 — Equilibrium	Ionic equilibrium — buffer pH (Henderson-Hasselbalch)	[HX] = [X ⁻] → pH = pK _a ; use given K _b of conjugate base to find K _a and hence pK _a = 14 – pK _b .
72	B	14 — Some Basic Principles of Organic Chemistry	IUPAC nomenclature of branched alkanes	Longest chain = 7 C (heptane); substituents at C3 (ethyl) and C5 (methyl) → 3-ethyl-5-methylheptane.
73	D	16 — Organic Compounds Containing Halogens	Substitution reactions of alkyl halides	C ₂ H ₅ Cl + AgCN → ethyl isocyanide (C ₂ H ₅ NC, foul-smelling) — ambident nucleophile gives –NC with AgCN.
74	D	5 — Solutions	Raoult's law — negative deviation, intermolecular forces	Chloroform-acetone form H-bonds between H of CHCl ₃ and lone pair on acetone's O → stronger A-B attraction → negative deviation.
75	C	5 — Solutions	Concentration units — molality	m = moles solute / kg solvent; 2.5 g acetic acid (M=60) in 75 g benzene → m ≈ 0.555 mol/kg.
76	C	10 — P-Block Elements	Group 13 elements — BCl ₃ monomer vs AlCl ₃ dimer (Al ₂ Cl ₆)	Evaluate statements about ECl ₃ (E = B or Al): BCl ₃ is monomeric (electron deficient); AlCl ₃ dimerises via Cl bridges.
77	C	3 — Chemical Bonding and Molecular Structure	Resonance and formal charge	Ozone (O ₃): central O (+1), terminal single-bonded O (–1), terminal double-bonded O (0).
78	C	4 — Chemical Thermodynamics	First law of thermodynamics — ΔU = Q – W	Heat absorbed by system = +500 J; work done by system = +200 J → ΔU = +300 J.
79	B	4 — Chemical Thermodynamics	Gibbs free energy; ΔG° = ΔH° – TΔS°; spontaneity	Compute ΔH° from ΔU° + Δn _g RT, then ΔG° = ΔH° – TΔS° at 298 K; sign of ΔG° → spontaneity.
80	B	18 — Organic Compounds Containing Nitrogen	Amines — reduction of nitriles to primary amines	LiAlH ₄ /H ₂ O, H ₂ /Ni, Na(Hg)/EtOH all reduce R–CN to R–CH ₂ –NH ₂ (primary amine); Sn/HCl and Br ₂ /aq.NaOH do not.
81	C	12 — Co-ordination Compounds	Ligands — ambidentate ligand	Ambidentate ligand can bond through two different donor atoms (e.g., SCN ⁻ via S or N; NO ₂ ⁻ via N or O) — not oxalate or EDTA.
82	B	2 — Atomic Structure	Photon energy E = hv; photoelectric / radiation calculations	Photons per second = (power converted to light) / (energy per photon) = (150 W × 0.08) / (hv).
83	C	10 — P-Block Elements	Group 15 (N, P, As) — bonding (pπ-pπ in N ₂ ; dπ-dπ back-bonding for P, As)	Identify the incorrect statement among bonding characteristics of nitrogen family.
84	A	9 — Classification of Elements and Periodicity in Properties	Periodic trends — metallic character	Metallic character decreases across a period and increases down a group; rank Na, Be, P, Mg, Si accordingly.
85	B	6 — Equilibrium	Solubility / K _{sp} -type equilibrium calculation	BiO(OH)(s) ⇌ BiO ⁺ + OH ⁻ ; K = [BiO ⁺][OH ⁻] = x ² = 4 × 10 ⁻¹¹ → [OH ⁻] = 2 × 10 ⁻⁶ M.

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86	D	3 — Chemical Bonding and Molecular Structure	Hybridization; sigma / pi bond count	Match C ₂ H ₄ (5σ+1π), C ₂ H ₂ (3σ+2π), CH ₄ (4σ), NH ₃ (3σ) to their bonding descriptors.
87	D	6 — Equilibrium	K _p vs K _c relationship	$K_p = K_c(RT)^{\Delta n_g}$; use stoichiometry to find Δn_g and compute K _p from given K _c .
88	D	17 — Organic Compounds Containing Oxygen	Alcohols — conversion to alkyl halide, then dehydrohalogenation to alkene	Propanol + PCl ₅ → propyl chloride (X = POCl ₃); + alc. KOH → propene (Y); identify intermediates and products.
89	D	15 — Hydrocarbons	Alkanes — free-radical chlorination (UV)	C ₂ H ₆ + Cl ₂ (UV) → C ₂ H ₅ Cl (X); subsequent reactions in sequence to identify final product Z.
90	D	10 — P-Block Elements	Carbon family (Group 14) — reaction of CO ₂ with hot coke	CO ₂ (g) + C(s) → 2CO(g): 1 volume reactant → 2 volumes product → net increase in gas volume.

Notes on borderline / cross-listed items

A few questions sit at the boundary of two syllabus units. These are the choices made and why:

Q54 (V₂O₅ / Fe catalysts)

Although catalysis itself sits in Unit 8 (Chemical Kinetics), the catalysts in question are transition-metal oxides / metals used in industrial inorganic processes (Contact, Haber), so the question is classified under Unit 11 (d- and f-Block Elements).

Q59 (steam reforming of methane)

Methane is an alkane, so the reaction is placed under Unit 15 (Hydrocarbons). It also has ties to industrial hydrogen production, but the syllabus has no separate Hydrogen unit — the alkane-reactivity framing is the primary fit.

Q60 (multi-step organic conversion)

Mixes aromatic sulfonation (Unit 15) with carboxylic-acid → alcohol reduction (Unit 17). Classified under Unit 17 because most of the reagent matching concerns functional-group transformations on oxygen-containing compounds.

Q73 (ethyl isocyanide formation)

Could be tagged under Unit 18 (nitrogen compounds, isocyanide product), but the underlying reaction is a nucleophilic substitution on an alkyl halide (C₂H₅Cl), so Unit 16 (halogen compounds) is the more accurate placement.

Q82 (photons per second)

Touches Unit 17 of Physics syllabus, but in Chemistry the same content is taught under electromagnetic radiation / $E = h\nu$ in Unit 2 (Atomic Structure).

Q88 (alcohol → alkyl chloride → alkene)

Starts with an alcohol, so primary classification is Unit 17 (Oxygen). The dehydrohalogenation step is in Unit 15 (Hydrocarbons), so the question crosses both units.

Q90 (CO₂ + coke)

Stoichiometric in flavour but the chemistry concerns the carbon family and CO/CO₂ — placed under Unit 10 (P-Block Elements).